RECIPENET: RECIPE QUALITY PREDICTION AND SEMANTIC CLUSTERING

Apply residual neural networks to tabular data for mapping semantic similarity in latent space

Ian Auger - CS 615

The Problem

Recipe Search

- Recipe search engines rely heavily on text-relevance to return filtered results to Users
- Search relevance can miss quality signals:
  - Taste
  - Ease
  - Reliability
- User review skew overwhelmingly positive and are noisy, making differentiation between similar recipes difficult
- Text relevance does not equate to recipe quality

Goal:

Train neural networks that can:

- Predict recipe quality scores
- Learn dense embeddings representing recipe similarity

These embeddings can later support:

- recommendation
- clustering
- semantic re-ranking in search systems

Reference Work

Deep Residual Learning for Image Recognition

He et al., 2016

- Foundational paper focusing on construction of residual connections allowing neural networks to learn the residual mappings instead of full transformations, enabling deeper models to train successfully
- Addresses the degradation problem where deeper networks become harder to optimize, while allowing gradients to propagate more easily through deep architecture

Revisiting Deep Learning Models for Tabular Data

Gorishniy et al., 2021

- This paper evaluates modern neural architectures for tabular datasets and demonstrates that simple residual MLP architectures can be strong baselines.
- Demonstrates that ResNet-style architectures perform competitively on tabular tasks by employing residual blocks that improve training stability

Raw Data

Food.com Recipes and Interactions

Recipes Table (231k rows)

- **name** — recipe title
- **ingredients** — list of ingredient strings
- **steps** — list of instruction strings
- **description** — recipe summary
- **tags** — categorical labels (cuisine, diet, holiday)
- **nutrition** — list of 7 numeric values
- **minutes** — prep time
- **n_steps** / **n_ingredients** — structural metadata

Interactions Table (1.1M rows)

- **user_id** — anonymous user
- **recipe_id** — recipe identifier
- **rating** — 1–5 stars
- **review** — free-text review

Prior Work - MLC Pipeline

Upstream Pipeline (Pre-Modeling Phase)

This project builds on a prior pipeline that transforms raw Food.com data into cleaned, structured features, and converts free-text reviews into consistent multi-label signals.

Key steps:

- **Data Cleaning & Normalization**
  - Standardized ingredient and instruction text
  - Removed noise and inconsistencies in review text
  - Structured recipe-level metadata into tabular format

- **Multi-Label Classification**
  - Defined a 17-tag taxonomy capturing themes in user feedback
  - Generated a labeled dataset (~65K reviews) using a zero-shot DistilBERT model
  - Learned tag-specific centroids in embedding space
  - Embedded reviews using Word2Vec representations
  - Assigned tags based on distance to centroids with calibrated thresholds

Ingestion Data

Recipe Table

- Recipe metadata: name, description, tags
- Processed text fields: ingredients, steps
- Structural features:
  - cooking time (minutes)
  - number of steps / ingredients

Nutritional features:

- calories, fat, sugar, sodium
- protein, carbs, saturated fat

Reviews Table

- user_id, recipe_id, date
- rating
- cleaned review text

Learned Tag Signals

- Similarity scores (sim_*)
  - continuous relevance to each tag
  - derived from embedding distance to tag centroids
- Predicted labels (pred_*)
  - binary indicators (0/1)
  - thresholded from similarity scores

Target Variable

Problem: Rating Distribution Issues

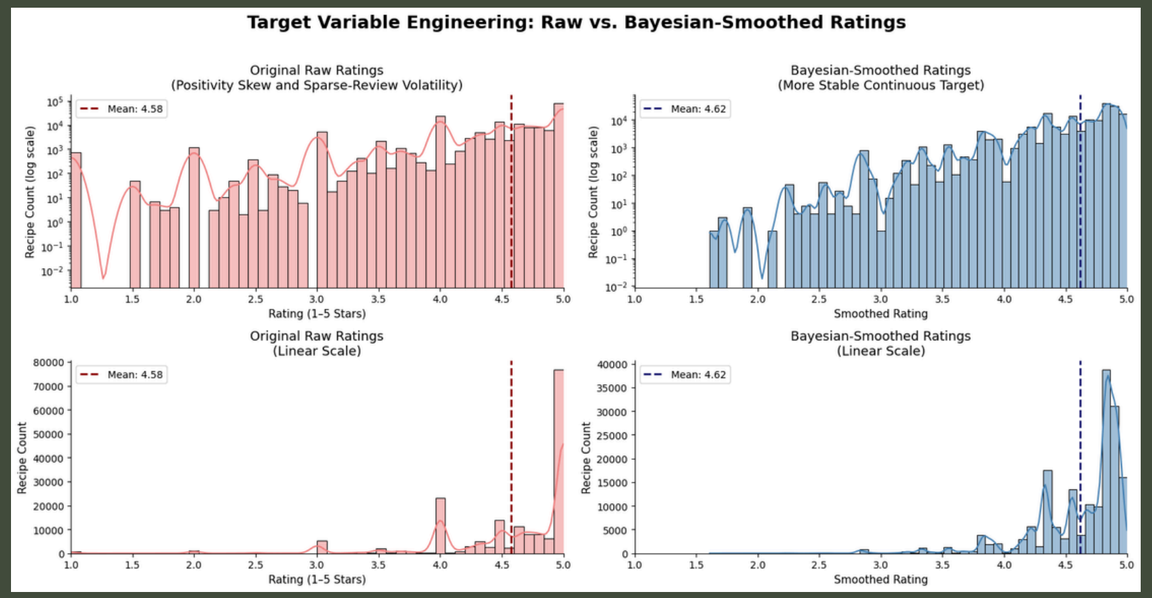
- Ratings are heavily skewed toward high values (4–5 stars)
- Low variance reduces model's ability to differentiate recipe quality
- Recipes with few reviews exhibit high volatility

Solution: Bayesian Smoothing

- Adjust ratings using:
  - global mean
  - review count
- Shrinks extreme values toward a stable estimate

Impact

- Produces a more continuous and stable target
- Reduces noise from low-sample recipes
- Improves ranking resolution between similar recipes



Feature Engineering

Recipe Metadata

- Structured attributes (ingredients, tags, time, nutrition)
- Multi-label encoding with top-N frequency filtering

Review-Derived Signals

- Aggregated to recipe level
- Tag frequency: % of reviews with signal
- Tag intensity: avg similarity when present

Target Engineering

- Bayesian-smoothed rating
- Stabilizes low-review recipes
- Improves ranking differentiation

Feature Scaling

- Standardization + MinMax normalization
- Ensures stable optimization across heterogeneous features

Architecture Overview

Dual-Stream Modeling Strategy

Instead of utilizing a single encoder, the architecture implements a dual encoder approach, treating each group of data as distinct and allowing the model to learn specialized representations for each. Then the groups are fused into a shared latent space for prediction

Recipe Metadata

- structure: minutes, steps, ingredients
- nutrition features
- recipe tags / ingredient indicators

Semantic Review Signals

- aggregated review tag prevalence
- aggregated tag intensity and prediction features
- learned qualitative feedback patterns
    -

Architecture Variation

Controlled Architecture Comparison

All models use the same:

- input preprocessing
- train/validation/test setup
- Bayesian-smoothed target

Only the prediction architecture changes.

Models Tested

Shallow MLP

Simple feedforward baseline on fused features.

Deep MLP

Adds depth to capture more complex nonlinear interactions.

Residual

Adds skip connections to improve gradient flow and feature reuse.

Residual_V2

Residual blocks + normalization for more stable optimization and smoother embeddings.

Residual_V3

Experimental gated residual variant with more expressive block transformations.

Two-Tower

Keeps metadata and review encoders separate until late fusion.

Experimental Framework

Model Testing

To evaluate model performance I ran each model through a controlled experiment matrix to isolate the impact of architcutre on performance and represntation quality.

Controlled Setup

- Fixed preprocessing pipeline and feature set
- Temporal train / validation / test split
- Bayesian-smoothed rating target

Architecture Sweep

- Shallow, Deep, Residual, Residual V2/V3, Two-Tower
- Consistent hidden dimensions and training budget

Feature Ablation

- All Features (metadata + review tags)
- Metadata Only
- Tag Only

Loss Functions

- MSE (baseline regression)
- Huber (robust to outliers)
- Log-Cosh (smooth alternative to MAE/MSE)

Training Diagnostics

- Training / validation loss curves
- Gradient norm tracking (stability signal)
- Early stopping + LR scheduling

Evaluation

- Test RMSE (primary)
- MAE / MSE (secondary)
- Full-corpus embedding extraction for downstream analysis

Model Evaluation

Feature Signal Dominates Performance

- Best models (all features): RMSE ≈ 0.296–0.298
- Tag-only models remain competitive (RMSE ≈ 0.30–0.31)
- Metadata-only models degrade significantly (RMSE ≈ 0.37+)

Semantic Review Features Are Critical

- Review-derived tags provide the majority of predictive signal
- Recipe metadata alone lacks sufficient resolution to differentiate recipe quality

Architecture Differences Are Marginal

- Performance spread across architectures is < 0.003 RMSE
- No clear winner based on predictive accuracy alone

Loss Function Effects Are Model-Specific

- No universally dominant loss function across architectures
- Huber and MSE both perform well depending on configuration

Key Takeaway

- Feature engineering > architecture complexity for this task
- Model selection should prioritize representation quality, not just RMSE

Architecture	Loss Function	RMSE
deep	mse	0.2964
shallow	huber	0.2964
residual_v2	mse	0.2973
residual	huber	0.2975
two_tower	huber	0.2980
residual_v3	huber	0.2997

Representation Quality Across Architectures

Deep models collapse representation space

- Embeddings form narrow, trajectory-like structures
- Indicates learning of a near 1D rating axis

Residual & Two-Tower preserve geometry

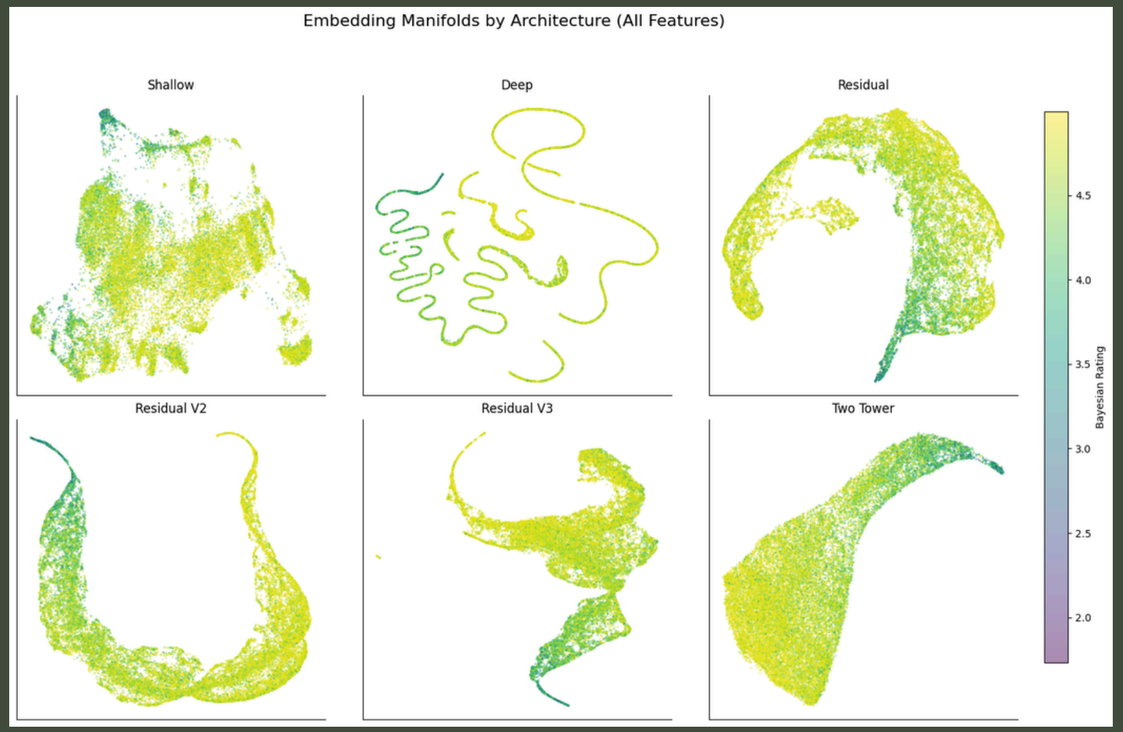
- Smoother, continuous manifolds
- Stronger local neighborhood structure
- Better clustering of semantically similar recipes

Implication for Retrieval

- Embedding quality directly impacts semantic search
- Structured latent spaces enable meaningful similarity comparisons

Models selected for downstream use

- Residual V2
- Two-Tower



Selecting the Production Architecture

Final Candidates

- Residual V2
- Two-Tower

Predictive Performance

- RMSE:
  - Residual V2: 0.2973
  - Two-Tower: 0.2980
- Spearman correlation:
  - Residual V2: 0.508
  - Two-Tower: 0.505
- Both provide strong ranking signal and predictive accuracy

Key Differentiator: Representation Quality

- Residual V2:
  - Smoother, more coherent embedding space
  - Stronger local neighborhood preservation
- Two-Tower:
  - Good separation, but less consistent structure

Decision Criteria

- Downstream use: semantic retrieval + re-ranking
- Embedding geometry prioritized over marginal metric differences

Final Selection

- Residual V2 chosen as production model

Final Embedding Space (Residual V2)

Continuous Latent Structure

- Large, smooth manifold across the recipe space
- No hard clustering — representation is continuous

Quality-Aligned Geometry

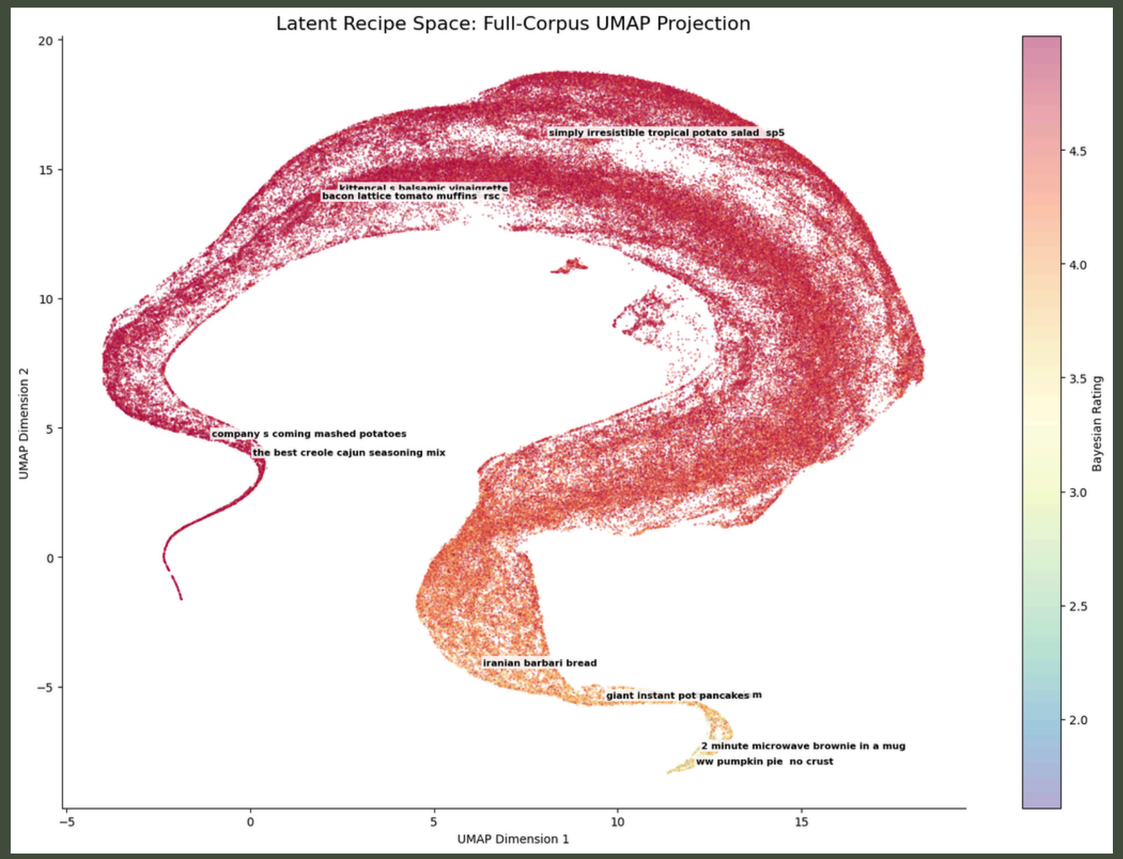
- High-rated recipes cluster along outer arc
- Lower-rated recipes concentrated in lower branch
- Smooth gradient indicates learned ranking signal

Semantic Neighborhoods

- Nearby recipes share similar characteristics
- Embedding space preserves local similarity structure

Implication for Retrieval

- Enables semantic similarity + re-ranking
- Supports ranking beyond keyword matching



Model Limitations

Systematic Bias in Predictions

- Model consistently overestimates low-rated recipes
- Strong compression toward high rating range (4–5)

Error Distribution

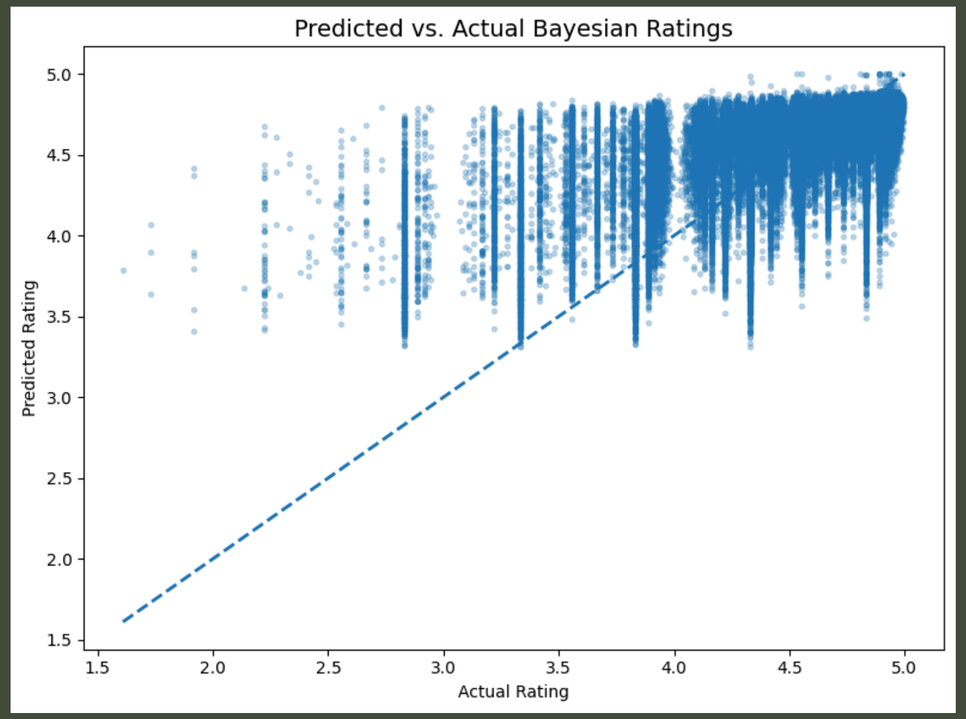
- Residuals centered near 0 for high-rated recipes
- Large negative residuals for low-rated recipes

Root Cause: Data Imbalance

- Dataset heavily skewed toward positive ratings
- Limited signal for distinguishing poor-quality recipes

Implication

- Model is strong for ranking good vs great
- Less reliable for identifying truly low-quality recipes



Conclusion

Architecture vs. Representation

- Similar predictive performance across all models
- Substantial differences in learned embedding geometry

Key Finding

- Residual architectures produce smoother, more coherent latent spaces
- Better preservation of local neighborhood structure

Final Model Selection

- Residual V2
- Strong ranking signal and high-quality embedding structure

Outcome

- Learned a continuous semantic representation of recipe quality and similarity

Application

- Enables semantic re-ranking in the retrieval pipeline
- Improves search beyond keyword matching

Thank You!